

obtain a 3D view. 3D spectacles try to achieve this same effect from one flat image source. Polarising spectacles have lenses that filters out images projected in one colour. The left lens may allow only a red image and the right one may allow only a blue image. Combining these two generates a greyscale three-dimensional view. Active spectacles have a shutter over each lens that alternately open and shut in coordination with an alternating image on screen, to generate a flicker-prone image. This is often available as an accessory to your video card, but rarely used because it's quite a headache for very unimpressive performance.

However, things have been improving. There is a new 3D display in town and it uses a toy technology to give a believable field of depth, requiring no equipment to be worn by the viewer. StereoGraphics is already selling such displays as SynthaGram monitors. Applications for SynthaGrams need to be developed using a special Software Development Kit provided by the company. Philips Research has prototype monitors that show almost three-dimensional images, from any normal application. They use special algorithms to interpret and enhance depth in 2D images.

The technology used to create these auto-stereoscopic displays is called Lenticular Display, and is a sophisticated form of something that has fascinated us all as children. In any stationary shop, you can find rulers that have a ribbed plastic surface on one side, applied over two images. Depending upon your angle of view, one image or the other is revealed. This effect has been known and used for nearly a century, but only recently has it been refined enough to be used in 3D displays. The lenses are angled to project two slightly different images, simultaneously. The angle on the lenses is such that one image is visible to the left eye while the other is visible to the right eye. The juxtaposition of these two images gives a 3D appearance to the combined view.

The computer display is formed with a combination of an LCD screen with a lenticular screen. The display is driven by any video card that has dual independent monitor support. The lenticular screen is a sheet with small lenses arrayed edge to edge in precise alignment with pixels on the underlying display. The display driver lights up appropriate pixels to push out two images at once to the viewer. Of

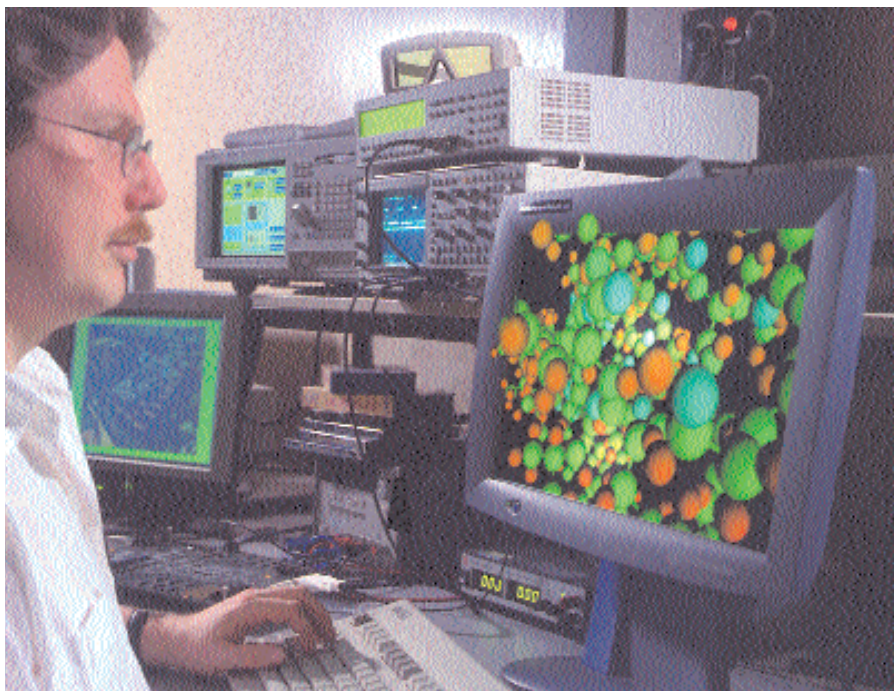


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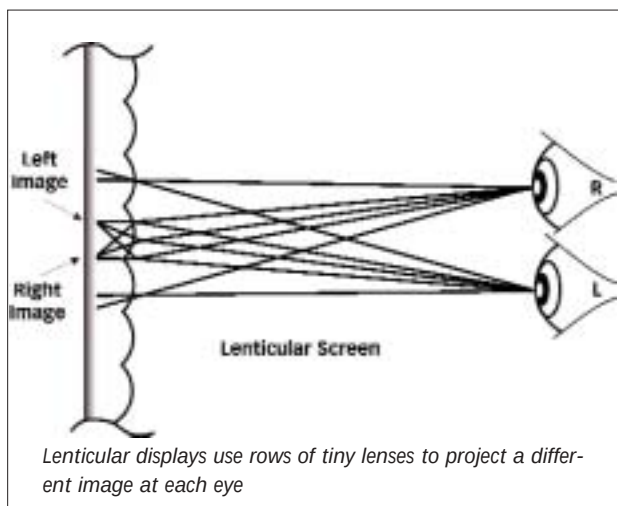
A prototype Philips monitor showing pseudo-3D images

course, because of the nature of the display, as you move your point of view around the display, the two images may not be visible separately. To compensate for this, SynthaGram monitors have nine viewing zones, i.e. areas where your angle of view is optimal for the images to appear in 3D. Elsewhere you will see a flat image much like a regular display.

Applications for such 3D displays are many, ranging from presentation of scientific and research data, to business intelligence to playing *DOOM 4* in realistic horror. But the most interesting opportunities with such displays is not just the specific niche applications, but the possibility of a real, usable three-dimensional interface that literally adds another dimension to computing.

OLED and beyond

The LCD is king when it comes to sharp, flat and thin screen displays, but there are challengers to the throne, none more likely to succeed than OLED screens. DuPont recently launched its version of OLED as the Olight division. Kodak has been working on this technology for over a decade



Lenticular displays use rows of tiny lenses to project a different image at each eye

after its researchers came up with the earliest breakthroughs, and such companies as Universal Display are busy collecting portfolios of patents for licensing to manufacturers.

Organic Light Emitting Diodes (OLED) are composed of complex organic compounds sandwiched between two electrically conducting materials. When a small current is applied to the conductors, the organic compound emits light. This effect is called electro luminescence. OLED construction is similar to an LCD in that it is a multilayer display. Active matrix OLEDs use transistors to control the light level of individual pixels in an OLED just like an LCD display. However, unlike an LCD, an OLED display does not necessarily have a top layer of glass.